Question 2: Economic impact of an ageing population

Extract 5: The costs and benefits of an ageing population

Table 2: Singapore: Population statistics (thousand)

	1990	2000	2010	2020
Total population	3047	4028	5077	5686
Resident population	2736	3273	3772	4044
0 – 19 years	851	929	918	803
20 – 64 years	1720	2109	2515	2626
65 years & over	164	235	338	614
of which 80 years & over	27	40	69	124
Old-age dependency ratio*		11	13	

Note: Figures may not sum due to rounding

*Old-age dependency ratio: Residents aged 65 years & over **per hundred** residents aged 20–64 years

Source: Singapore Department of Statistics, June 2023

An ageing population and slower labour force growth affect different economies in many ways. For example, the growth of GDP slows, working-age people pay more in tax to support the elderly, and government budgets are put under strain due to the burden of increased costs of health and retirement programmes for old people.

An ageing population puts budgetary pressure on the government and society because the support ratio, which is the ratio of workers to consumers, declines. Between 2015 and 2050 the support ratio is expected to drop by 0.26% a year in the US, 0.40% in other high-income nations, and 0.82% in China.

Public sector transfer payments in the form of pensions, health care, and long-term care pose a greater problem as populations age. These payments, even after subtracting the portion funded by tax payments from the elderly, account for a large portion of government budgets. Projections indicate that typically, these programmes will become unaffordable unless, for example, productivity is increased, taxes are raised or benefits for the elderly are reduced, such as by increasing the retirement age.

In general, the higher the proportion of their consumption that the elderly finance themselves, the less cost falls on working adults as higher taxes. Unlike the US, where people retire later and rely more on their own assets in old age, Europe has a heavy reliance on public sector transfer payments to pay for consumption by the elderly. When older people contribute little to their own consumption, either through asset income or continuing to work, it puts a heavy cost on the labour force as the population ages.

Firms may choose to cut investment in the domestic economy substantially, even as interest rates fall, if they think that output and consumption growth will slow in response to an ageing population. Should firms become pessimistic, even if central banks drive interest rates below zero, the economy could have zero growth with high unemployment for many years. This may be shown in Japan's virtually flat economic growth in recent decades and Europe's failure to recover from the global financial crisis.

An ageing population will mean more old-age dependency. But it may also bring more capital per worker leading to rising productivity and wages. Whether an ageing population is good or bad, there are no simple answers.

Source: Cost of Ageing, Finance & Development, March 2017, Vol. 54, No. 1

Extract 6: Why population ageing matters: A global perspective

For the first time in history, people aged 65 and over now outnumber children under age 5 (as shown in Figure 2). In 2006, almost 500 million people were 65 and older, accounting for about 8% of the world's population.

By 2030, the world is predicted to have 1 billion older people, accounting for 13% of the total population. While proportions of older people have typically been highest in high-income countries, the most rapid increases in older populations are now occurring in low-income countries. Between 2006 and 2030, the number of older people in low-income countries was projected to increase by 140% as compared to an increase of 51% in high-income countries.

Figure 2: Young children (age < 5) and older people (age 65+) as a percentage of global population

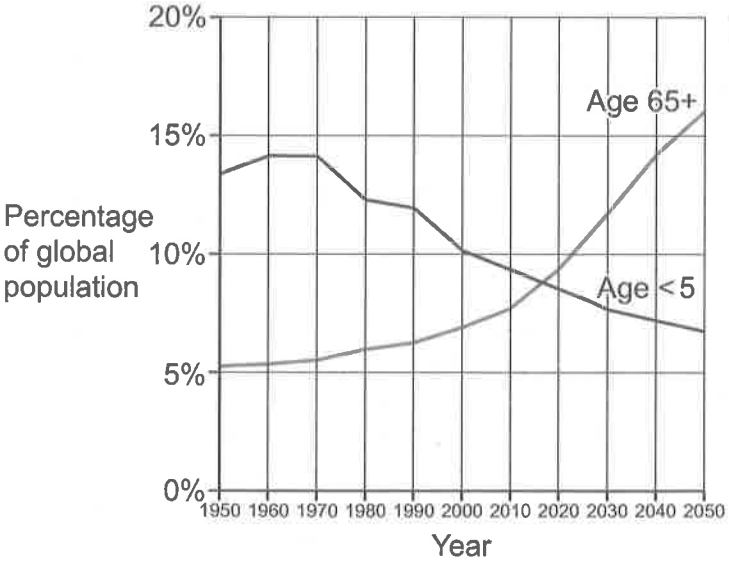
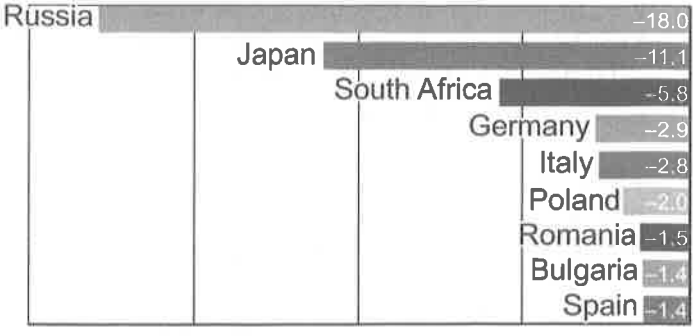


Figure 3: Projected population decline of selected countries between 2006 and 2030 (millions)



More than 20 countries were projected to experience population decline by 2030. As Figure 3 shows, for example, Russia's population was expected to shrink by 18 million between 2006 and 2030, a decrease of nearly 13%.

Source: adapted from Why Population Aging Matters: A Global Perspective, USA National Institute on Aging, March 2007

Extract 7: Singapore's approach to its ageing population

An economy with an ageing population can benefit from an accumulation of wealth and savings as older individuals save for consumption in old age and invest in their human capital, especially in health and education. If governments are forward-looking and act early, encouraging such behaviour, this can potentially lead to improved productivity over longer lifespans and an increase in gross national income.

Singapore has one of the most rapidly ageing populations in the world with a life expectancy of around 83 years. Its government has invested significantly in life-long learning initiatives to boost the country's human capital potential as well as to promote personal development and social integration.

SkillsFuture, a national programme launched in 2014, provides every Singaporean aged 25 and above with an opening credit of S$500 that they may use to attend approved skills-based courses. It also offers work-study programmes to strengthen the links between industry and education for students and mid-career professionals.

The Central Provident Fund Lifelong Income for the Elderly (CPF LIFE) scheme provides coverage for the retirement income needs of Singaporeans. It is financed through compulsory monthly contributions and provides a monthly payout during retirement.

Source: These countries are most ready to deal with ageing populations, World Economic Forum, 18 Feb 2020

Extract 8: More migrant workers needed to offset ageing population

An International Monetary Fund (IMF) report has stated that high-income countries such as the UK, the US and Japan risk being overwhelmed by their ageing populations. It calls on them to throw open their borders to more migrant workers in response. However, net annual migration of European Union (EU) nationals to the UK in 2017, for example, fell to its lowest level for five years. UK companies have raised growing concerns about labour shortages.

Within the next few decades, working-age adults will need to support double the number of elderly people than they do now, putting immense pressure on welfare systems and reducing potential economic output by as much as 3% by 2050.

The IMF called for more investment in education to get more people into work and to increase the chances of workers staying in a job beyond retirement age in order to increase the labour supply. Shifting cultural norms and government policies have already started this process by increasing the number of women entering and remaining in the workforce. The IMF stated, however, that more women and older workers alone may not be enough for economies to solve the problem. It said labour market participation would still eventually decline and was also likely to remain below current levels, even if more women worked and more older people stayed in work for a longer time period.

Immigration policies allow countries to control the influx of people, such as migrant workers, who want to establish residence within their borders. They include rules regarding rights of access to the country and permission to participate in the labour market.

The IMF report concluded that: 'Unless technology delivers offsetting productivity gains, these findings highlight the need for many high-income countries to rethink immigration policies to boost their labour supply, alongside policies to encourage older workers to postpone retirement.'

Source: adapted from The Guardian, 9 April 2018

(a) With reference to Extract 5:

(i) calculate and compare the old-age dependency ratio for Singapore in 1990 with that in 2020. [3]

(ii) explain **one** reason for the change in the old-age dependency ratio for Singapore between 1990 and 2020. [2]

(b) Explain how the changes in the populations of countries shown in Figure 3 might affect aggregate supply and aggregate demand in those countries. [4]

(c) With reference to Extract 5, explain why firms may choose 'to cut investment in the domestic economy substantially, even as interest rates fall'. [3]

(d) Discuss whether the benefits to an economy of having an ageing population outweigh the costs. [8]

(e) Discuss whether immigration is the best way of promoting economic growth in an economy with an ageing population. [10]

[Total: 30]